

Basic feasible solutions

Kent Andersen

In this note we consider a polyhedron:

$$P = \{x \in \mathbb{R}^n : Ax \leq b\},$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and m is the number of inequalities. The set $M := \{1, \dots, m\}$ is used to index the inequalities $Ax \leq b$. The rows of A are denoted $a_{i.} \in \mathbb{R}^n$ for $i \in M$. Given any $y \in \mathbb{R}^n$, the set $I(y) := \{i \in M : a_{i.}^T y = b_i\}$ is the index set for the inequalities satisfied with equality by y .

If a point $\bar{x} \in P$ is the only solution of the system $a_{i.}^T x = b_i$ for $i \in I(\bar{x})$, then $\bar{x}$ is called a *basic feasible solution* of $Ax \leq b$.

Definition 1 (*Basic feasible solutions*)

If $\bar{x} \in P$ is the only solution to the system:

$$a_{i.}^T x = b_i \text{ for } i \in I(\bar{x}), \tag{1}$$

then $\bar{x}$ is called a basic feasible solution of $Ax \leq b$. ■

Extreme points of P are points $\bar{x} \in P$ that are not convex combinations of points in $P \setminus \{\bar{x}\}$. We will show that basic feasible solutions are exactly the extreme points of P .

Theorem 1 (*Extreme points are equivalent to basic feasible solutions*)

$\bar{x} \in P$ is extreme for $P \iff \bar{x}$ is a basic feasible solution for $Ax \leq b$. ■

Theorem 1 gives an algorithm for checking whether a point $\bar{x} \in \mathbb{R}^n$ is extreme for P : If $A\bar{x} \leq b$ holds, then $\bar{x}$ is extreme for P if and only if $\bar{x}$ is the only solution to (1). Solving (1) will reveal whether $\bar{x}$ is the only solution.

In the remainder we prove Theorem 1. We prove each direction separately in Theorem 2 and Theorem 3 below:

Theorem 2 (*Basic feasible solutions are extreme points*)

If $\bar{x} \in P$ is not extreme for P , then $\bar{x}$ is not basic feasible for $Ax \leq b$. ■

Proof: Suppose $\bar{x} \in P$ is not an extreme point of P . Hence $\bar{x}$ is a convex combination of two points in $P \setminus \{\bar{x}\}$ (Theorem 2 in note 2.(A)), i.e., one may write $\bar{x} = \lambda \cdot x^1 + (1 - \lambda) \cdot x^2$, where $x^1, x^2 \in P \setminus \{\bar{x}\}$ and $\lambda \in]0, 1[$. Since $a_i^T x^1 \leq b_i$ and $a_i^T x^2 \leq b_i$ holds for all $i \in I(\bar{x})$ one has:

$$b_i = a_i^T \bar{x} = \lambda \cdot (a_i^T x^1) + (1 - \lambda) \cdot (a_i^T x^2) \leq b_i \quad \text{for all } i \in I(\bar{x}),$$

which implies $a_i^T x^1 = a_i^T x^2 = b_i$ for all $i \in I(\bar{x})$. Hence x^1 and x^2 also solve (1), i.e., $\bar{x}$ is not the only solution to (1). By definition this shows $\bar{x}$ is not a basic feasible solution for $Ax \leq b$. ■

We finally show extreme points for P are also basic feasible for $Ax \leq b$:

Theorem 3 (*Extreme points are basic feasible*)

If $\bar{x} \in P$ is not basic feasible for $Ax \leq b$, then $\bar{x}$ is not extreme for P . ■

Proof: Suppose $\bar{x} \in P$ is not a basic feasible solution for $Ax \leq b$. Let $\bar{y} \neq \bar{x}$ be an alternative solution to (1), and let $\bar{d} := \bar{y} - \bar{x} \neq 0_n$ be the direction that leads from $\bar{x}$ to $\bar{y}$.

We show there exists $\alpha^* > 0$ such that the line segment between $\bar{x} + \alpha^* \cdot \bar{d}$ and $\bar{x} - \alpha^* \cdot \bar{d}$ lies in P . Since $\bar{x}$ lies in the middle of this line segment, i.e., one has $\bar{x} = \frac{1}{2} \cdot (\bar{x} + \alpha^* \cdot \bar{d}) + \frac{1}{2} \cdot (\bar{x} - \alpha^* \cdot \bar{d})$, this shows $\bar{x}$ is not an extreme point for P . Hence we now show there exists $\alpha^* > 0$ such that $a_i^T (\bar{x} + \alpha \cdot \bar{d}) \leq b_i$ holds for all $i \in M$ and $\alpha \in [-\alpha^*, \alpha^*]$.

For every $i \in M$ satisfying $a_i^T \bar{d} \neq 0$ let $\alpha_i := \frac{b_i - a_i^T \bar{x}}{|a_i^T \bar{d}|}$. The question is which values $\alpha \in \mathbb{R}$ satisfy all inequalities $a_i^T (\bar{x} + \alpha \cdot \bar{d}) \leq b_i$ for $i \in M$. For every $i \in M$ there are three cases:

- (i) $a_i^T \bar{d} = 0$: $a_i^T (\bar{x} + \alpha \cdot \bar{d}) \leq b_i$ holds for all $\alpha \in \mathbb{R}$.
- (ii) $a_i^T \bar{d} < 0$: $a_i^T (\bar{x} + \alpha \cdot \bar{d}) \leq b_i \iff \alpha \geq -\alpha_i$.
- (iii) $a_i^T \bar{d} > 0$: $a_i^T (\bar{x} + \alpha \cdot \bar{d}) \leq b_i \iff \alpha \leq \alpha_i$.

Finally observe that one has $\alpha_i > 0$ for all $i \in M$ satisfying $a_i^T \bar{d} \neq 0$. Indeed, for every $i \in M$ with $a_i^T \bar{d} \neq 0$ one has $i \notin I(\bar{x})$ and $b_i - a_i^T \bar{x} > 0$ (since $a_i^T \bar{d} = a_i^T \bar{y} - a_i^T \bar{x} = 0$ holds when $i \in I(\bar{x})$), and therefore $\alpha_i = \frac{b_i - a_i^T \bar{x}}{|a_i^T \bar{d}|} > 0$.

It follows from (i)-(iii) above that the value $\alpha^* := \min\{\alpha_i : a_i^T \bar{d} \neq 0\} > 0$ has the property that $a_i^T (\bar{x} + \alpha \cdot \bar{d}) \leq b_i$ holds for all $i \in M$ and $\alpha \in [-\alpha^*, \alpha^*]$, which shows $\bar{x} + \alpha^* \cdot \bar{d} \in P$ and $\bar{x} - \alpha^* \cdot \bar{d} \in P$. ■